

5(a)(i)	<i>any three from:</i> addition of, <u>activated</u> / phosphorylated, nucleotides ; A free nucleotides ref. to complementary, nucleotides / bases / strands ; forms phosphodiester bonds (between adjacent nucleotides) ; adds nucleotides to, elongating / growing, polynucleotide / strand / AW ; ref. to proofreading ability of DNA polymerase ;	3
---------	--	----------

Question	Answer	Marks
5(a)(ii)	any two from: ref. to leading strand <u>and</u> lagging strand (in context of movement of DNA polymerase along each strand in opposite directions) ; ref. to (Okazaki) fragments for lagging strand ; (lagging strand) requires DNA ligase ; AVP ; e.g. DNA polymerase can only add nucleotides to the, 3 prime / 3', end of the growing DNA strand	2
5(b)	any two from: 1 cell, cycle/regulation, checks, prevent tumour cell progressing to mitosis ; 2 stop cell cycle in G₂ phase / at end of S phase ; 3 ref. to rapidly dividing (tumour) cells will be most affected by etoposide ; 4 ref to consequence of, cuts / damage, in DNA ; e.g. may lead to tumour cell, death / apoptosis 5 DNA polymerase prevented from moving along strands so replication of tumour cell cannot occur ; 6 AVP ; e.g. can be attached to monoclonal antibodies (specific to tumour cells)	2

Question	Answer	Marks
----------	--------	-------